Course Content	
CONTINUOUS AND DISCRETE TIME SIGNALS AND SYSTEMS Continuous Time (CT) &Discrete Time (DT) signals- Classification - standard signals – basic operations on signals - Continuous time and discrete time systems - properties - Linear Time Invariant (LTI) systems - Stability - Causality - Convolution Integral, Convolution sum.	**16 Hours**

The Language of Machines: Introducing the Protocol

An Analogy for Communication

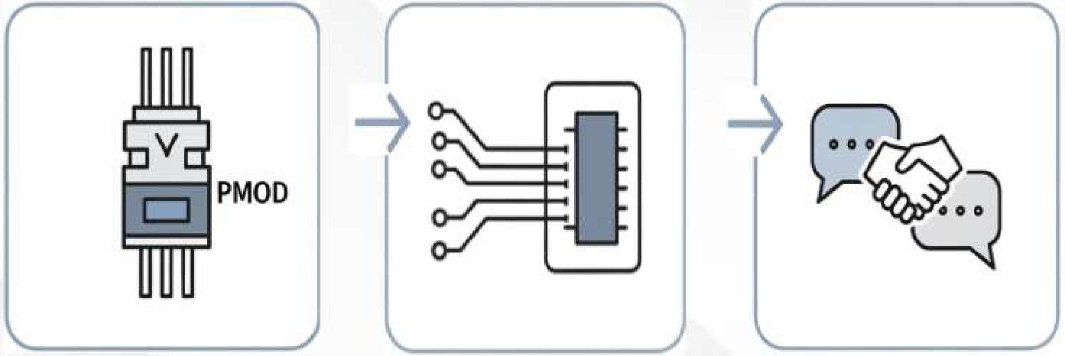
- **Interface (PMOD, Grove)**: The physical meeting room.
- **Bus (Wires)**: The infrastructure (e.g., table and chairs).
- **Protocol (SPI, I2C)**: The shared language and rules of etiquette.

Formal Definition:

A system of rules that allows two or more entities of a communications system to transmit and receive information .

Core Function:

To transform a chaotic stream of raw electrical signals into structured, reliable, and meaningful information.



INTERFACE:
The Meeting Room

BUS
The Table

PROTOCOL:
The Shared Language & Rules

The Three Pillars of a Protocol

Communication Protocol

Please click the buttons to learn more

Semantics (The Vocabulary):

- Defines the meaning and interpretation of the data.
- This approach accelerates development by allowing rapid connection to a vast ecosystem of peripherals.

The Three Pillars of a Protocol

Communication Protocol

Please click the buttons to learn more

Timing (The Rhythm):

✕

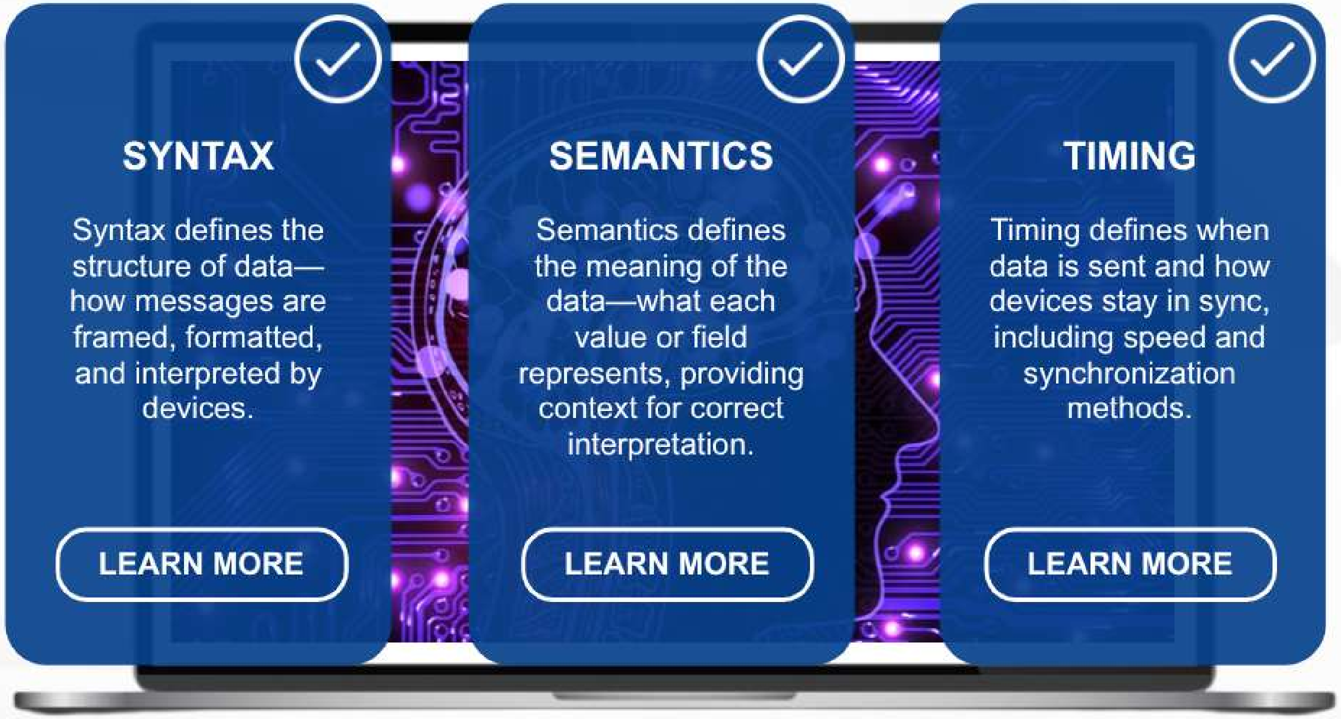
- Defines when to send data and how fast.
- **Examples:** Data rate (baud rate), synchronization method (clock signals), turn-around times.

→

The Three Pillars of a Protocol

Communication Protocol

Please click the buttons to learn more



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SYNTAX

Syntax defines the structure of data—how messages are framed, formatted, and interpreted by devices.

LEARN MORE

✓

SEMANTICS

Semantics defines the meaning of the data—what each value or field represents, providing context for correct interpretation.

LEARN MORE

✓

TIMING

Timing defines when data is sent and how devices stay in sync, including speed and synchronization methods.

LEARN MORE

Serial vs. Parallel Communication

Core Concept

The fundamental choice between sending data one bit at a time (serial) or many bits at once (parallel).

Parallel Communication Analogy

- An 8-lane superhighway.
- 8 cars (bits) travel simultaneously.
- High throughput, but wide and expensive to build.

Serial Communication Analogy

- A single-lane road or railway track.
- Cars (bits) travel one after another.
- Narrower, cheaper, and better for long distances.

Parallel Communication: The Multi-Lane Superhighway

Definition:

The simultaneous transmission of multiple bits (e.g., a full byte) over multiple data lines, synchronized by a common clock signal.

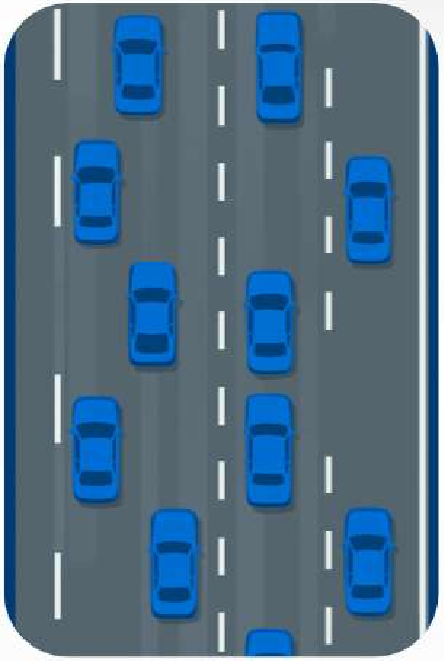
Primary Advantage: High Speed (Theoretically)

- At the same clock frequency, an 8-bit parallel bus is theoretically 8 times faster than a 1-bit serial bus.

Key Applications:

- **Legacy**: Old printer ports (IEEE-1284), IDE/PATA hard drives.
- **Modern (Short-distance, controlled):** High-speed internal buses like CPU-to-RAM (DDR SDRAM) and some LCD/Camera interfaces (MIPI) where the physical environment is tightly controlled. And Host interfaces like PCIe.

Parallel Communication



The Achilles' Heel of Parallel: Skew and Crosstalk

High Pin Count & Cost:

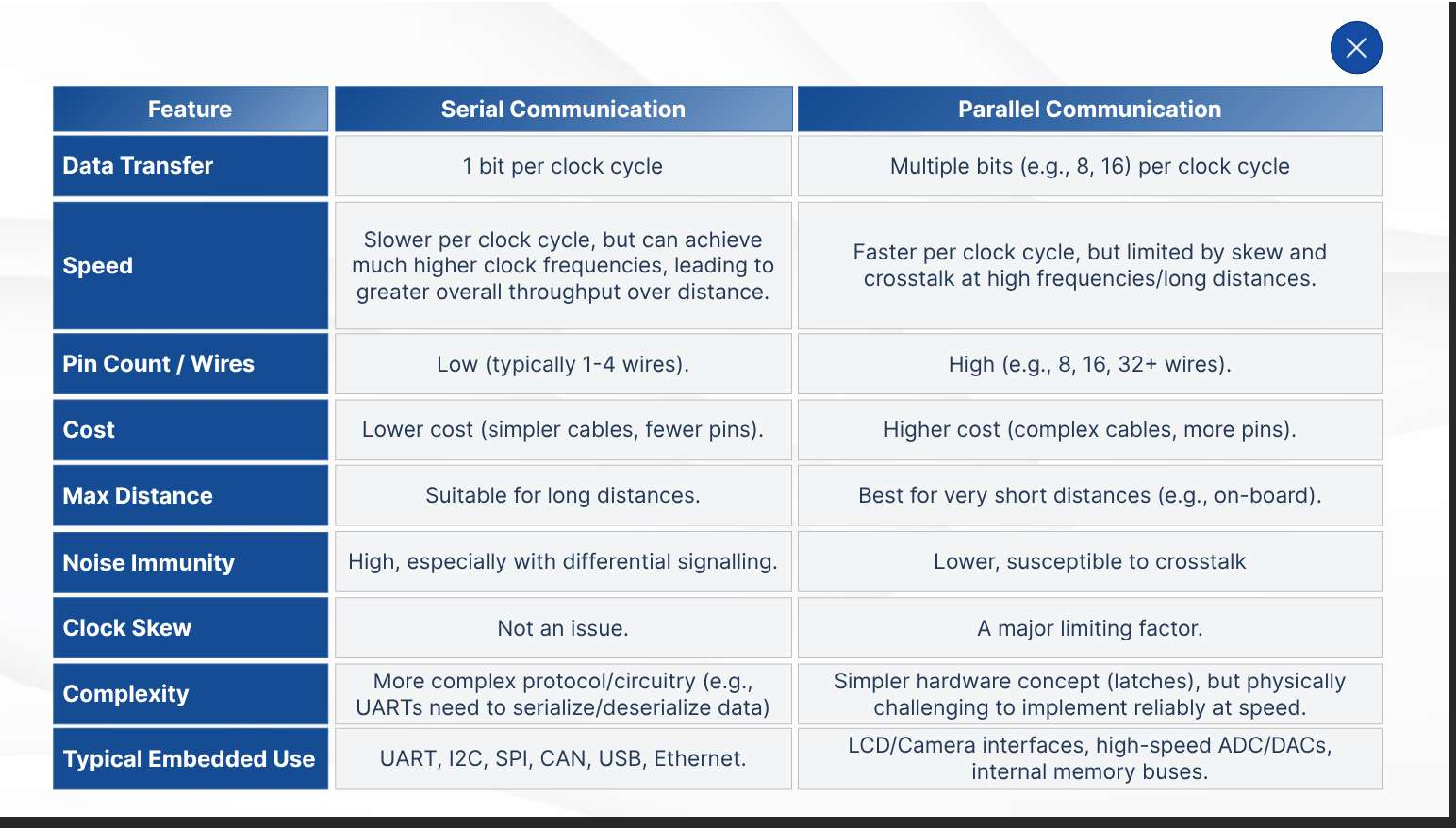
- Requires many MCU I/O pins.
- Cables are wider, more complex, and more expensive.

Crosstalk:

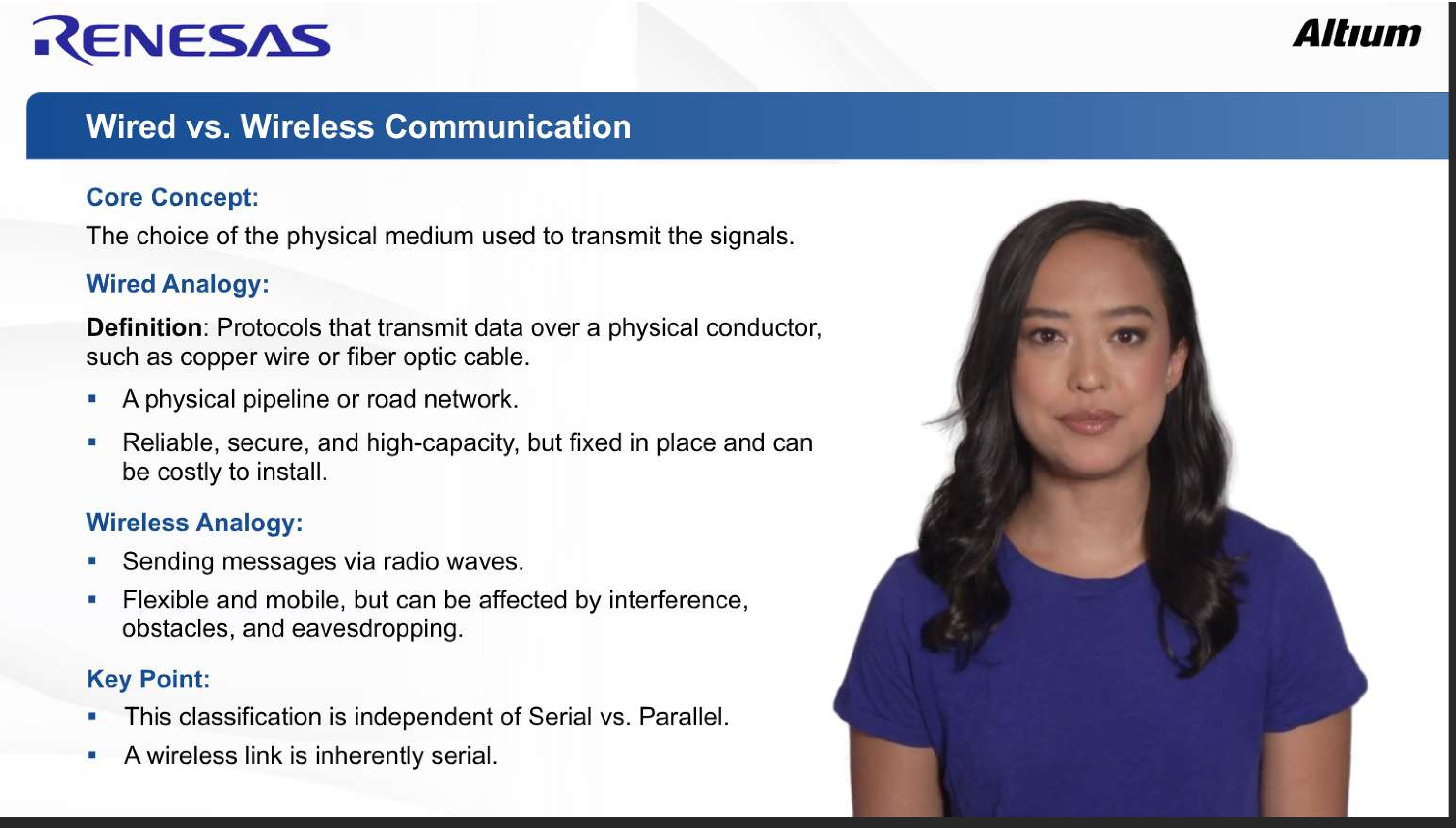
- Electromagnetic interference between adjacent parallel wires corrupts data.
- This problem worsens at high speeds and over longer distances.

Clock Skew (The Speed Killer):

- **Definition**: The phenomenon where bits sent at the same time from the transmitter arrive at the receiver at slightly different times.
- **Cause**: Tiny, unavoidable physical differences in the lengths and electrical properties of the parallel wires.
- **Effect**: At high clock speeds, the clock period is very short. If the skew is a significant fraction of the clock period, the receiver will latch the wrong data, causing catastrophic errors.



Feature	Serial Communication	Parallel Communication
Data Transfer	1 bit per clock cycle	Multiple bits (e.g., 8, 16) per clock cycle
Speed	Slower per clock cycle, but can achieve much higher clock frequencies, leading to greater overall throughput over distance.	Faster per clock cycle, but limited by skew and crosstalk at high frequencies/long distances.
Pin Count / Wires	Low (typically 1-4 wires).	High (e.g., 8, 16, 32+ wires).
Cost	Lower cost (simpler cables, fewer pins).	Higher cost (complex cables, more pins).
Max Distance	Suitable for long distances.	Best for very short distances (e.g., on-board).
Noise Immunity	High, especially with differential signalling.	Lower, susceptible to crosstalk
Clock Skew	Not an issue.	A major limiting factor.
Complexity	More complex protocol/circuitry (e.g., UARTs need to serialize/deserialize data)	Simpler hardware concept (latches), but physically challenging to implement reliably at speed.
Typical Embedded Use	UART, I2C, SPI, CAN, USB, Ethernet.	LCD/Camera interfaces, high-speed ADC/DACs, internal memory buses.



Wired vs. Wireless Communication

Core Concept:

The choice of the physical medium used to transmit the signals.

Wired Analogy:

Definition: Protocols that transmit data over a physical conductor, such as copper wire or fiber optic cable.

- A physical pipeline or road network.
- Reliable, secure, and high-capacity, but fixed in place and can be costly to install.

Wireless Analogy:

- Sending messages via radio waves.
- Flexible and mobile, but can be affected by interference, obstacles, and eavesdropping.

Key Point:

- This classification is independent of Serial vs. Parallel.
- A wireless link is inherently serial.

Wired vs. Wireless: A Comparative Analysis

Factor	Wired Communication	Wireless Communication
Speed	Typically higher and more consistent.	Variable; depends on signal strength, interference, and standard.
Reliability	Very high. Not affected by RF interference.	Lower. Susceptible to interference from other devices, walls, etc.
Security	High. Requires physical access to the network.	Lower. Signals are broadcast; requires strong encryption.
Mobility	Low. Devices are physically tethered.	High. The primary advantage.
Installation	More complex and costly; requires physical cabling.	Simpler and often cheaper; no cables to run.
Latency	Low and predictable.	Higher and more variable.

Conclusion: The choice is application-driven. Wired is preferred for stationary applications requiring maximum speed, reliability, and security (e.g., industrial control, server rooms). Wireless is essential for applications where mobility and ease of installation are paramount (e.g., IoT, mobile devices).

A 2X2 Matrix Classification

SERIAL	**WIRED/SERIAL** i2C UART SPI CAN Ethernet	**WIRELESS/SERIAL** BLE WiFi Zigbee zIE Cellular (LTE/5G)
PARALLELL	DDR Memory Bus Legacy Printer Port LCD/Camera Interfaces Wired	5G NR with MIMO Wi-Fi 6/7 with MU-MIMO UWB Wireless VR/AR streaming Satellite Internet Wireless

Our Roadmap: The Protocols We Will Master

1 On-Chip & Board-Level Serial

- SPI (Serial Peripheral Interface)
- I2C (Inter-Integrated Circuit)
- UART (Universal Asynchronous Receiver-Transmitter)

2 Robust Wired Networking

CAN

- CAN (Controller Area Network)

3 Wireless & IoT Protocols

BLE Wi Fi

- BLE (Bluetooth Low Energy)
- Wi-Fi

4 System-Level & PC Connectivity

USB
Ethernet

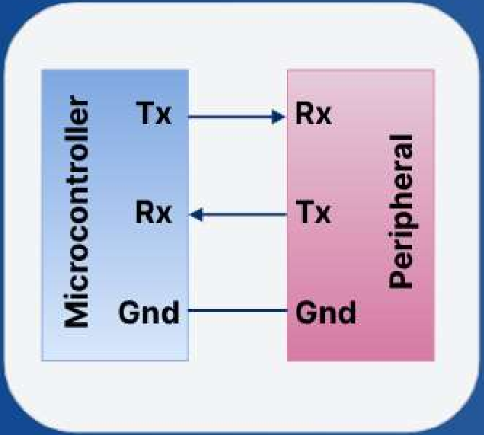
- USB (Universal Serial Bus)
- Ethernet)

"With these foundations, we are ready to master the rules, data frames, and real-world implementation of each of these powerful protocols."

The Workhorses: On-Chip & Board-Level Serial Protocols

Synchronous (SPI, I2C) vs. Asynchronous (UART).

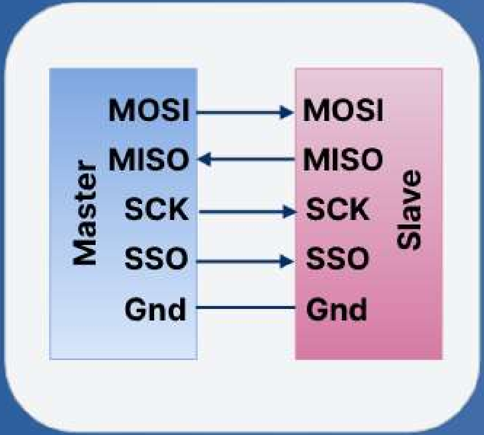
UART


Microcontroller
Tx → Rx
Rx ← Tx
Gnd — Gnd
Peripheral

UART
(Universal Asynchronous Receiver-Transmitter)

Simple, two-wire, point-to-point communication.

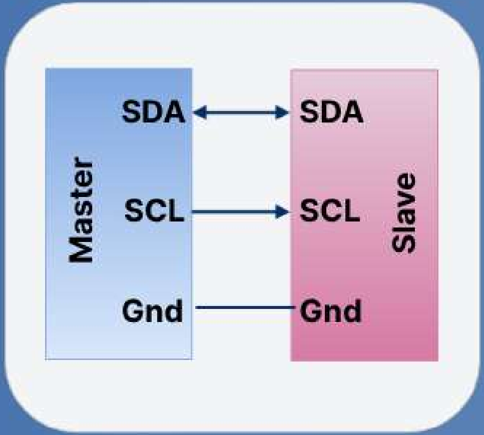
SPI


Master
MOSI → MOSI
MISO ← MISO
SCK → SCK
SSO → SSO
Gnd — Gnd
Slave

SPI
(Serial Peripheral Interface)

Fast, synchronous communication with a master-slave architecture.

I2C


Master
SDA ↔ SDA
SCL → SCL
Gnd — Gnd
Slave

I2C
(Inter-Integrated Circuit)

Efficient, two-wire bus for connecting multiple devices.